جامعة مصر للمعلوماتية
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Computer and Information Systems
Data Analysis Course

Data Analysis: Impact of Discounts on E-commerce Sales Performance

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Introduction

In an era where online shopping is accelerating in popularity, the pricing strategies deployed by e-commerce businesses become among the most impactful decisions they make. Discounts and promotional offers have become pervasive across online retail platforms. However, despite their widespread adoption, the real effect of discounts on sales volume, revenue and profitability remains a topic of considerable complexity.

The intuition behind discounting is straightforward: lower prices attract a higher number of buyers along with greater purchase volumes. Though, the real-world dynamics are more nuanced. Discount strategies can stimulate demand, reduce business costs and cultivate customer loyalty, but they can equally erode profit margins, attract price-sensitive consumers that leave at full price and reduce the perceived value of a product or service.

This report presents a rigorous, data-driven analysis and investigation into how discounts may or may not influence customer purchasing behavior and the performance of an e-commerce business. A dataset of 34,500 sample transaction records and 17 features is utilized for the process of exploring the relationship between discount depth and its impact on business performance.

By the end of this report, the reader should have a comprehensive understanding of the relationship between discounting and e-commerce sales and profitability, supported by descriptive statistics, data visualizations and hypothesis testing.

Research Question

How do discounts affect e-commerce sales and customer purchasing behavior, and to what extent do they impact profitability?

Hypothesis

Null Hypothesis (H_0): There is no statistically significant difference in sales amounts between discounted and non-discounted products.

Alternative Hypothesis (H_1): Discounts have a statistically significant positive impact on e-commerce sales.

Population of Interest:

The population of interest for this study is all e-commerce customers who order through online retail platforms.

Sampling Method:

This study uses a synthetic e-commerce dataset that simulates real-world online shopping transactions. The dataset used was generated to reflect realistic distributions of key variables such as pricing, discount rates, product categories and several customer demographics. As such, the sampling method utilized is best described as synthetic sampling, where different statistics were represented proportionally to their expected occurrence in real-world e-commerce environments.

The use of synthetic data is the most appropriate sampling approach for this study for several reasons. Firstly, it eliminates the legal and ethical complications of working with real transactional data from businesses. Moreover, it ensures complete data availability, with no missing values or restrictions, allowing for cleaner analysis of variable relationships without the complications that real-world datasets are typically faced with. The 34,500 records provide a sufficiently large sample size to ensure reliable results.

Bias Identification:

In any data analysis research and study, the identification of potential biases is fundamental to the validity of the findings. Several sources of bias could be considered for this particular analysis:

- **Synthetic Data Bias:** Since the dataset is synthetically generated, it may not fully capture the irregularities and non-linear patterns found in real data.
- **Category Representation:** Although the dataset contains multiple product categories, their relative proportions may cause a bias if one category is overrepresented.
- **Selection Bias in Discounting:** The dataset does not indicate which customers organically came across the discounts and which sought them. This distinction could matter for patterns in purchasing behavior.
- **Omitted Variable Bias:** Variables such as brand loyalty, seasonal effects and customer income are not included in the dataset. The lack of these variables may distort the final observed relationship between discounts and sales performance.

To mitigate the aforementioned biases, this analysis will rely on formal statistical testing methods, acknowledging the difference between correlation and causation when interpreting findings.

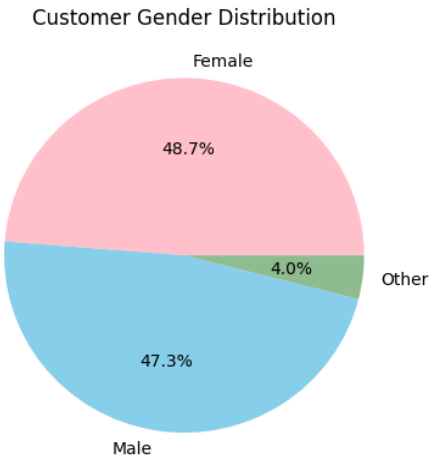
Dataset:

As previously mentioned, the dataset used in this analysis is a synthetic e-commerce sales dataset consisting of 34,500 transaction records across 18 features, with 1 feature added during the data preparation process. It was designed to simulate realistic online shopping data by encompassing a broad range of product categories, customer demographics, regions and pricing structures. The dataset spans to cover approximately a two-year window of commercial activity. A brief, comprehensive description of the dataset and its features can be found within the attached Colab Notebook.

Analysis:

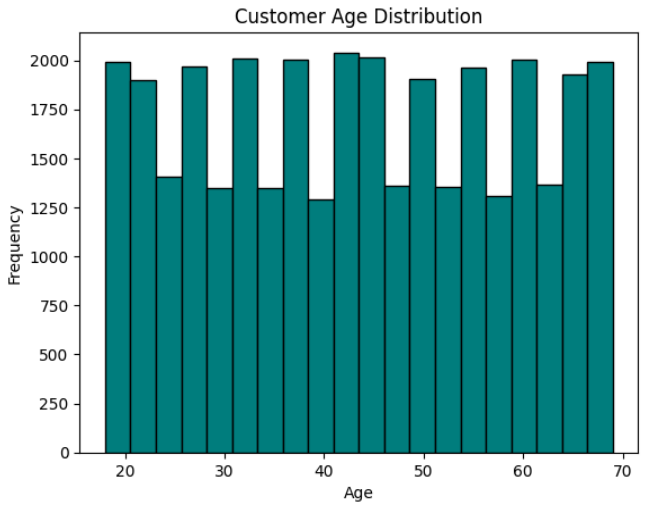
Section 1: Demographic Overview

Figure 1.1: Customer Gender Distribution



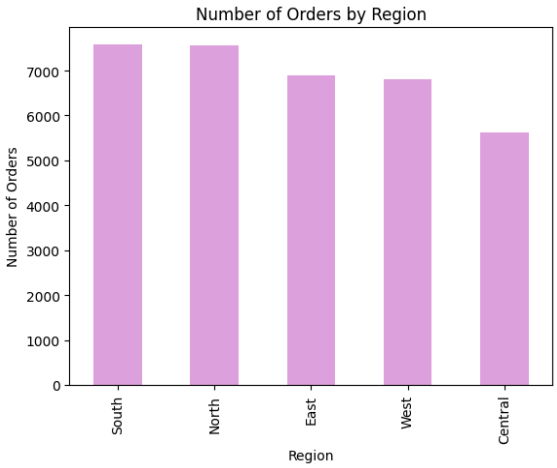
The pie chart illustrates a **largely balanced distribution** across three reported gender categories. This near-equal split indicated that the dataset does not skew toward a particular gender, which is critical for ensuring that the findings on purchasing behavior are not impacted by gender-based differences.

Figure 1.2: Customer Age Distribution



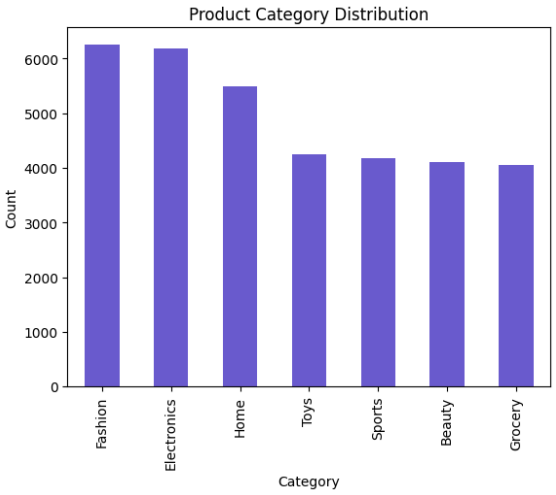
The histogram highlighting the customer age distribution reveals a broad, approximately **uniform** spread across the 18-69 age range, with no single age group being dramatically overrepresented. The **mean** customer age is about 43.5 years, suggesting a moderate-to-mature adult consumer base. Overall, the distribution indicates that e-commerce attracts customers from various life stages.

Figure 1.3: Orders by Region



The bar chart displaying the number of orders per region shows relatively **evenly distributed** numbers across all five regions, with no single region having a significantly larger share of total transactions.

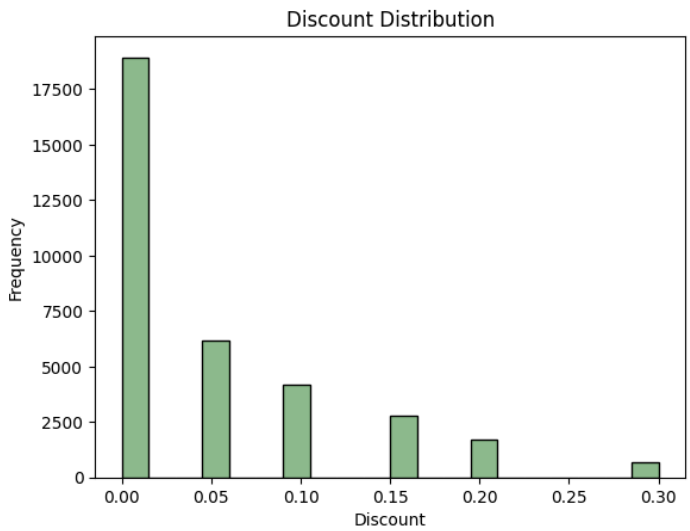
Figure 1.4: Product Category Distribution



The product category distribution bar chart reveals the frequency of orders across each category within the dataset. This composition is particularly relevant when examining discount effectiveness as discounting strategies are **unlikely to be uniformly effective** across all product types. A brief understanding of this baseline composition is fundamental for the category-level discount analysis in Section 4.

Section 2: Exploring Variables

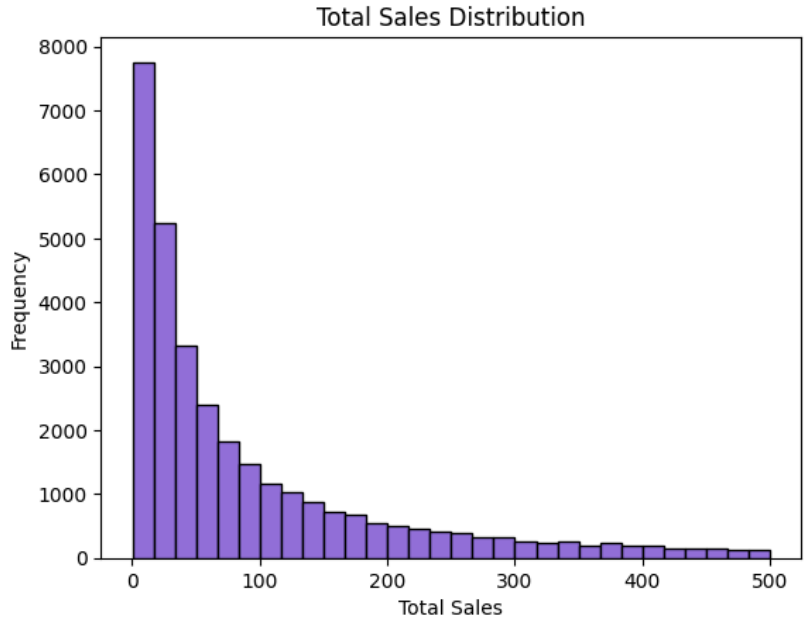
Figure 2.1: Discount Distribution



The discount distribution histogram reveals a **strongly right skewed distribution** with a large spike at the discount of 0, which is the **median** discount as observed from descriptive statistics. This confirms the fact that approximately half of all orders in the dataset are placed with no discount applied at all.

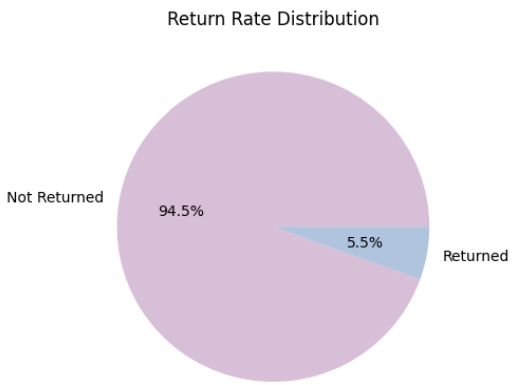
Moreover, the discounted orders are concentrated at **discrete intervals** with a **maximum** offer of 30% off. It is crucial to note that the discount variable is not truly continuous but rather takes on a small number of discrete values for this dataset. This discrete structure was accounted for and effectively utilized using a **has_discount** boolean variable to split the discount data into two groups for hypothesis testing: no discount (0.00) and discount.

Figure 2.2: Total Sales Amount Distribution



The histogram of total sales amounts, filtered to orders below 500 in order to improve visual clarity and reduce the influence of outliers, reveals a **strong right skewed distribution**. The vast majority of orders falls within modest ranges, the highest transaction frequencies concentrated in the lower bracket of sales. The **mean** total amount across all orders is approximately 170 while the **median** is about 57, indicating a significant pull from high-value outliers, such as the **maximum** (12,932).

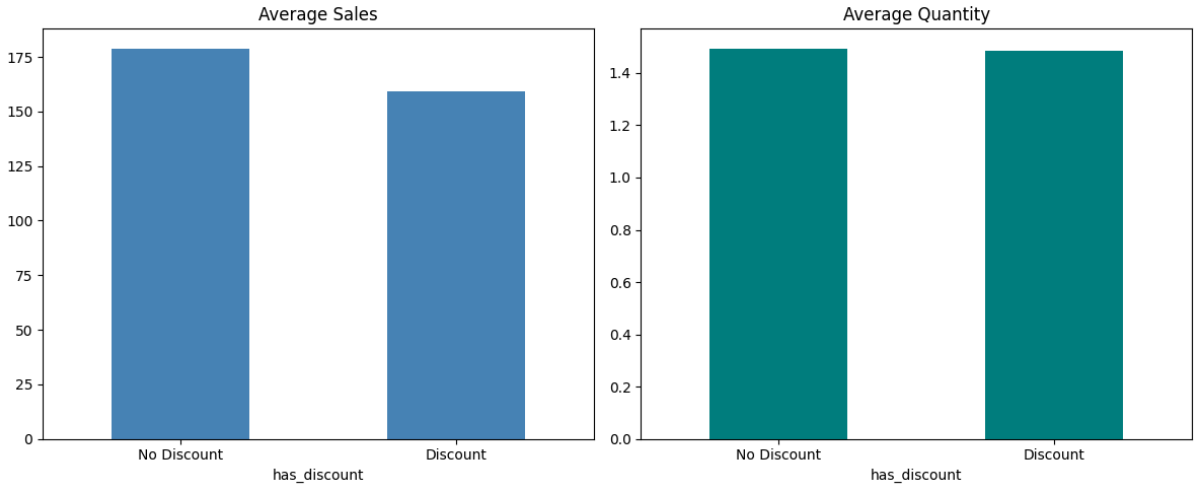
Figure 2.3: Return Rate Distribution



The return rate distribution pie chart shows that the overwhelming majority of orders are **not returned**. This relatively low return rate suggests that customers are generally satisfied with their purchases, or alternatively, that this synthetic dataset represents a market with limited awareness and assistance regarding return policies.

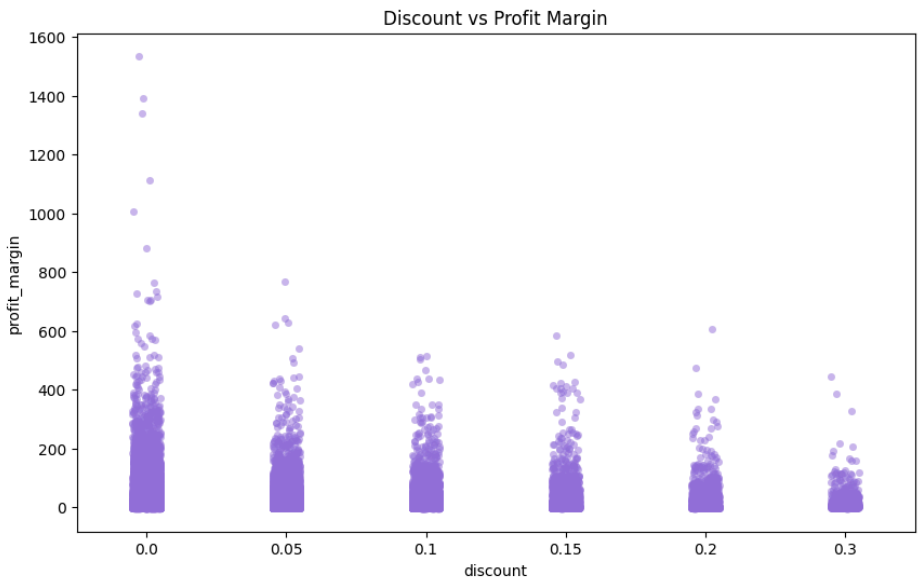
Section 3: Core Comparisons

Figure 3.1: Average Sales & Average Quantity by Discount



The dual bar chart presents a side-by-side comparison of the average total sales and the average quantity purchased for orders with and without a discount. Contrary to the common belief, the results reveal that **non-discounted** orders actually yield a somewhat **higher** average total sales amount for approximately equal quantities sold. It can be concluded that non-discounted products appear to produce slightly higher sales while being sold in the same amounts.

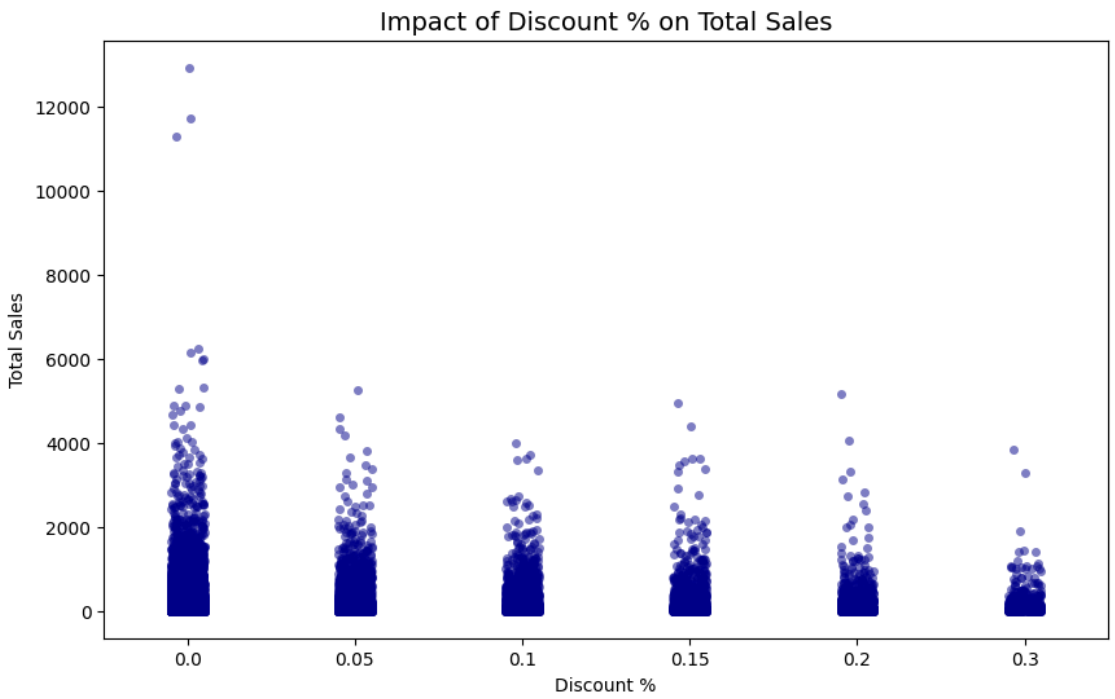
Figure 3.2: Discount vs. Profit Margin



The strip plot visualizing the relationship between the discount percentage and profit margin presents a revealing picture.

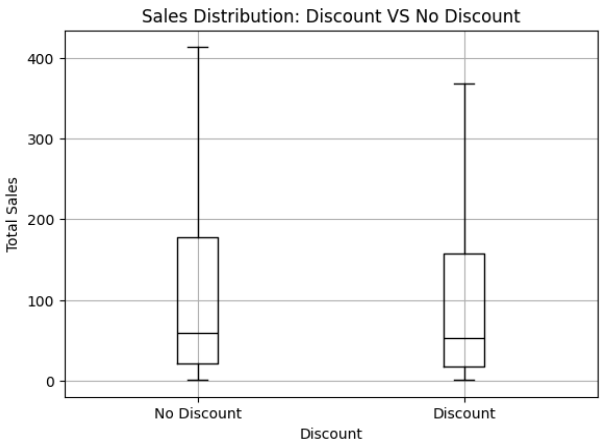
Across all discount tiers, including 0%, the profit margin values show considerable variability, but a **slight downward trend** is observable as discount percentage increases. This indicates, once again, that deeper discounts exert a measurable negative pressure on profitability. However, it is crucial to note that the relationship between discount depth and profitability is not linear, but complex.

Figure 3.3: Impact of Discount Percentage on Total Sales



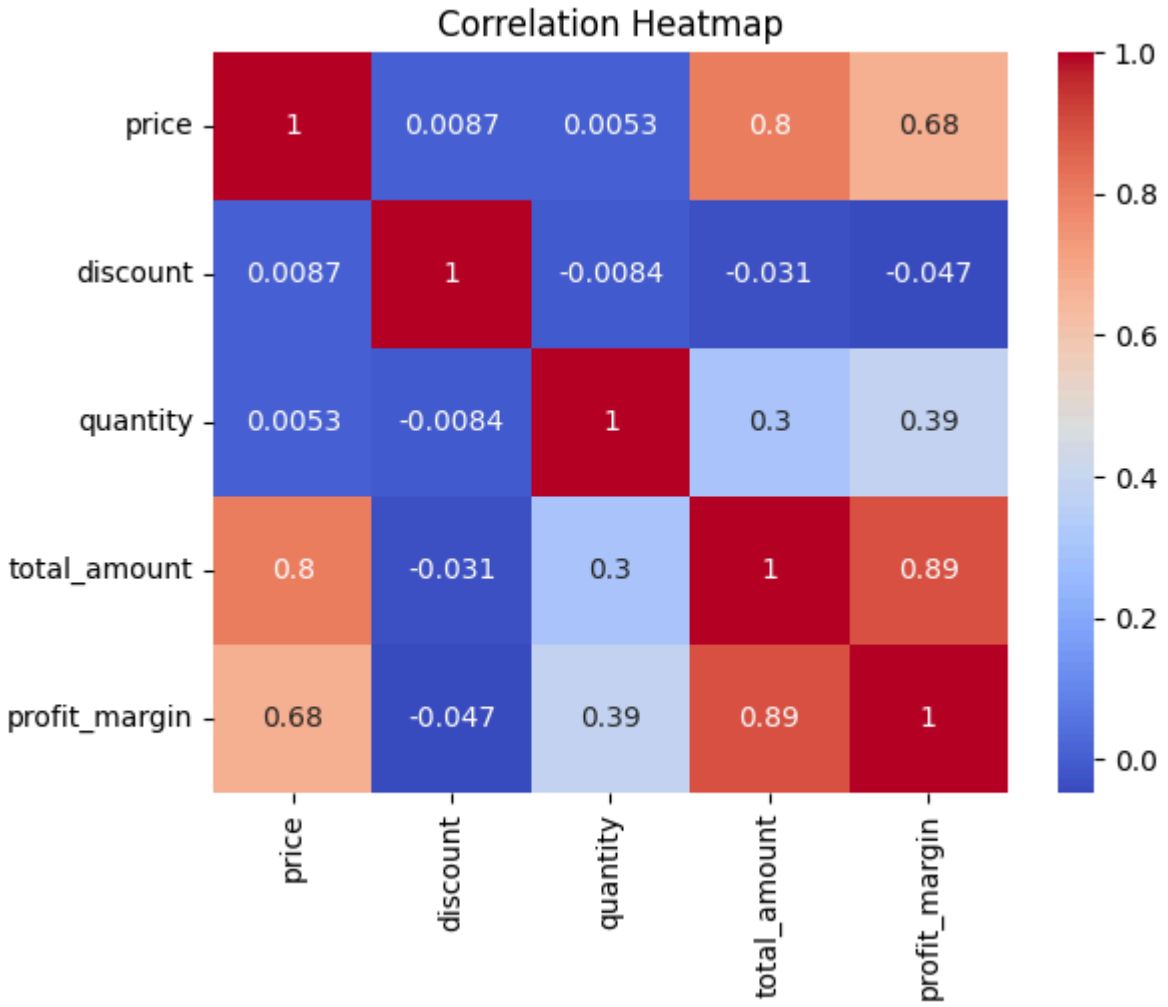
This scatter plot examines the impact of discount percentage on total sales. There is no clear relationship between discount depth and total sales amount, although another **subtle downward trend** may be observed, where the total sales at the 0% discount significantly exceed those at the highest discount percentage.

Figure 3.4: Sales Distribution by Discount



This boxplot comparing total sales distributions for discounted and non-discounted products reveals that the two groups are remarkably similar in their distributional shape, with a **nearly identical median** amount. This reinforces the finding from the dual bar chart: discounted orders do not appear to elevate sales amounts.

Figure 3.5: Correlation Heatmap



The correlation heatmap visualizes the correlation between pairs of the five key numeric variables significant to this study. The most prominent relationships, such as (price vs. total amount), (price vs. profit margin) and (profit margin vs. total amount) exhibit **strong, positive** correlations. These relationships are intuitive as higher-priced products naturally generate higher total order values.

It is revealed that discounts have a weak to negligible relationship with all other variables in the dataset.

Therefore, a consistent picture is painted: discounts have virtually no meaningful linear relationship with any key e-commerce business performance variables, and where a relationship does exist, it is slightly negative.

Hypothesis Testing Steps

Null Hypothesis (H_0): There is no statistically significant difference in sales amounts between discounted and non-discounted products.
Alternative Hypothesis (H_1): Discounts have a statistically significant positive impact on e-commerce sales.
Significance Level: $\alpha = 0.05$ (5%)

Step 1: Test Selection

The statistical test chosen for this analysis is **Welch's Independent Samples t-Test**. This test is appropriate for this analysis as the two groups under comparison (discount vs. no discount) are mutually exclusive and **independent**. Each transaction belongs to exactly one of the two groups. Additionally, the two groups cannot be assumed to have equal variances, and the sample size is extremely large, making the t-Test valid despite any skewness due to the **Central Limit Theorem**.

Step 2: Computations

Groups:

- **Group A:** All records where has_discount = 1 (*discounted*)
- **Group B:** All records where has_discount = 0 (*non-discounted*)

Test Statistic: -5.2177
P-value: 0.0000 (< 0.0001)

Step 3: Decision

The p-value is significantly lower than the alpha value. Therefore:

=> Reject the null hypothesis.

Conclusion

This study set out to investigate the complex relationship between discounts and e-commerce sales performance, specifically examining their impact on total sales, purchase quantity and profitability. Based on the conducted analysis and statistical hypothesis testing, the results indicate that discounts do not have a strong positive effect on sales as initially expected.

Across multiple visualizations, discounted and non-discounted orders demonstrated very similar sales patterns. In some cases, non-discounted transactions even exhibited slightly higher average sales amounts. Furthermore, the correlation analysis revealed that the relationship between discount percentage and key performance metrics such as total sales and quantity purchased is weak and, in some cases, slightly negative.

The analysis also highlighted a more consistent effect of discounts on profitability. As discount levels increased, profit margins showed a general downward trend, indicating that while discounts may influence pricing, it does not necessarily follow that revenue increases, or that purchasing activity improves.

The hypothesis testing results further support these observations. With a statistically significant p-value, the null hypothesis was rejected, confirming that there is sufficient evidence to support the existence of a difference between discounted and non-discounted transactions. However, the direction and practical significance of this difference indicate that discounts do not enhance sales performance in a beneficial or meaningful way.

These findings are particularly relevant in the context of the Egyptian market, where e-commerce is rapidly expanding and businesses are often relying on promotional offers like discounts as a competitive strategy. In such a price-sensitive market, it is commonly assumed that lowering prices will directly lead to increased sales. However, the results of this study suggest that that is not always the case as it could instead reduce profitability without significantly improving sales performance.

Understanding this relationship is important for online retail businesses operating in Egypt as it highlights the need for more strategic use of competitive techniques. Instead, companies may benefit from focusing on other factors such as product value, customer experience and branding in order to drive sustainable growth.

Overall, this study emphasizes that while discounts can be a useful tool in certain scenarios, they are not a guaranteed driver of sales success and should be applied with caution and strategic thinking.

Issues Faced

All significant problems faced during the process of conducting this study were the above mentioned discovered biases. No further problems were identified.